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Hierarchical Glicko-2 for Doubles Tennis

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Plagiarism Statement

The work contained in this thesis is my own work unless otherwise stated.

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Abstract

The International Tennis Federation (ITF), with support from Imperial College London, have developed a ranking system called the World Tennis Number (WTN). Every tennis player, regardless of their gender, age, nationality, or professional status, will have their skill ranked on a universal scale that ranges from 1 to 40. The algorithm underpinning the WTN is a hierarchical version of the Glicko-2 algorithm. The rating that is produced by the algorithm for a player is subsequently mapped so that it ranges from 1 to 40, which creates a player's WTN. By assigning players to groups and rating the groups as well as the players, the hierarchical version overcomes the issue of comparing players that play in homogeneous groups. Much of the work so far on the WTN has been designed with singles play in mind. The singles algorithm is inefficient when used on doubles play because doubles pairings change over time. Rating pairs of players would lose out on potentially important information, such as a player's results in a different pairing. This is why the concept of a notional opponent was developed in previous work. The notional opponent allows players involved in doubles matches to acquire an individual rating based on their doubles results. Thus we can compute a rating for any doubles pairing in the world, even if they have never played together before, by combining their rating using a skill aggregation function. In this report we make an amendment to the original proposed notional opponent. We then evaluate the concept of a notional opponent by comparing potential skill aggregation functions and extend it to the groups by creating notional groups. We assess the doubles algorithm by using factitious populations of players with skills that vary over time, as well as real data collected by the ITF.

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1. Introduction

In tennis there are 3 standard types of matches: singles, doubles, and mixed doubles. In terms of prize pools and global attention, the types of matches follow the above order. This results in some of the best tennis players in the world prioritizing singles tennis over doubles and mixed doubles. For example, the current men's singles world number 1, Novak Djokovic, is the joint most successful men's singles tennis player in the Open era with 20 Grand Slam men's singles wins, but he rarely competes in men's doubles tennis tournaments and has no Grand Slam men's doubles wins (ATP, 2021). Despite this, doubles and mixed doubles tennis is especially popular at the lower and more recreational levels of tennis. With 4 players on the court and less space to cover for each individual player, the game is more accessible to all types of players.

In this section we detail why the World Tennis Number (WTN) is needed, we summarise how the hierarchical Glicko-2 algorithm works, we discuss the aims and contributions made during this project, before outlining what is in the next chapters.

1.1. The Need for the WTN

In 2019 the International Tennis Federation (ITF) published a survey of worldwide tennis participation and performance. The key findings were that there were 87 million tennis players across 195 nations, with a gender balance of 53% male and 47% female (ITF, 2019). This report showed that tennis is a global game, immensely popular across all genders and age groups. The ITF is responsible for the organization and sanctioning of international competitions involving junior, professional, and senior players. For the ITF it is important that players are accurately ranked to improve the seeding in their events. Most tennis tournaments follow a knock-out style competition where the winner of each match continues to the next round until a champion is crowned. It is desirable for the best players to reach the latter rounds of these tournaments and the seeding of players can highly influence this. For example, the 2021 Australian Open men's singles final was played between the top seed, Novak Djokovic, and the number 4 seed, Daniil Medvedev. Without seeding, or even with poor seeding, these two players may have faced each other in an earlier round.

For the 211 national associations affiliated with the ITF, there are benefits of the WTN to be found at a more local level. The ranking system can be used to make more

appropriate and competitive matches. This will then increase participation from players because of the decrease in the number of one-sided matches. There is little to gain for the winning player when a match is too easy, while a heavy defeat can leave the losing player feeling embarrassed or falling out of love with the sport. While it is unlikely that the WTN will cause more players to start playing tennis, it is believed that those that do start will play for longer than the current average. These associations should also be able to quickly identify the very best players at junior level and adjust funding accordingly. If a player's individual rating combined with their group rating ranks them with players at a higher level, then they will stand out as deserving of investment.

The WTN is another tool for player evaluation, so it could be used in the selection process for high performance teams or college recruitment. Players can benefit from this since the WTN incorporates the results from all their matches, not just the high-profile events that are typically used to judge a player's talent. Players that may have missed an important tournament due to injury or suffered an unexpected early round loss can be reassured that their WTN will accurately reflect their skills. Players should also develop quicker since they can find opponents on a similar level using the WTN. Once in production the WTN will show live updates, so players will be able to track their progress and witness their number move up or down with each win and loss.

1.2. Hierarchical Glicko-2

The Glicko-2 algorithm (Glickman, 2001) is a method for assessing player strength in a game of skill, such as tennis. It was an adaptation to the Glicko algorithm (Glickman, 1999), by allowing the underlying model to have stochastic variance rather than a fixed variance. The Glicko-2 algorithm produces approximate Bayesian updates to the distribution of a player's strength¹. While Glicko-2 is capable of comparing many players at once and allows for player skill to exhibit sudden changes over time, it struggles with sparsity between players. The sparsity problem occurs when players play the majority of their matches in homogeneous groups, and very few against players from outside their group. Essentially there are not enough games played between players. In practice this is very common. For example, a veteran, male tennis player based in Ireland will likely only compete against other veteran, male tennis players based in Ireland. This makes it difficult to compare said player with someone from a different age group, gender, or country since there may be no matches played between the players or even common opponents. This issue makes it difficult for an algorithm that only uses match results to tell the difference between the best player in the world who dominates on the international stage and a very strong recreational player who plays against their friends every week.

The hierarchical variant of Glicko-2, developed at Imperial College London, offers a

¹The use of the words strength, skill and ability will be used interchangeably throughout to refer to the same idea.

solution to this problem. By clustering players into groups and assigning ratings to the groups, as well as the players, we can accurately rate players. While a player within a group may play very few games against opponents outside of their group, when we aggregate all the games played by players within a group against other groups, we can get a fairly accurate alignment between the groups. Once we know how one group compares against another group, we can adjust the players skill levels to compare them.

1.3. Project Aims

This report focuses on a specific part of a wider project that is being run by the ITF. The wider project is to produce a set of ratings that accommodates different levels of tennis competition on a single scale. This set of ratings is then transformed in to a WTN, which is a number that ranges from 1 to 40 and is rounded to two decimal places, e.g. a player may have a WTN of 25.53. Players with a similar WTN are expected to be close in skill and should have a competitive match against each other. Players that register to receive a WTN will be given both a singles number and a doubles number. The ITF have delegated some of the research and development of the project to students at Imperial College London. This includes how to automatically cluster players, determining the best way to rate players and transforming a player rating into a number. The research project undertaken in this report focuses solely on the algorithm that is used to rate players involved in doubles tennis. The goal is to extend the algorithm that is currently in place for singles tennis to doubles tennis.

1.4. Contributions

The specific contributions made to the WTN initiative in this report are summarised below.

1.4.1. Develop the concepts of a notional opponent

We adapt the Bayesian model for player skill to include a notional opponent and show how to adjust the Glicko-2 algorithm for different skill aggregation functions. Furthermore, we decompose a notional opponent into the player element, termed the ‘notional player’, and the group element, termed the ‘notional group’. We then apply the notional opponent idea to the groups, to allow a better alignment of the group strengths.

1.4.2. Evaluate different skill aggregation functions

We compare the performance of different skill aggregation functions, specifically the sum, mean, maximum and minimum, using simulations and real tennis data supplied by the ITF. We also show the hierarchical Glicko-2 algorithm outperforms the Glicko-2 algorithm when applied to the tennis data. We calculate the parameters of the distribution of a notional opponent's skill for the sum and mean aggregation functions.

1.5. Chapter Outline

Chapter 2 introduces some background information about the Glicko-2 algorithm before discussing the player sparsity problem that motivated the hierarchical variant of Glicko-2. We then discuss how the Bayesian model is adapted to include groups before highlighting some potential weaknesses of this methodology.

Chapter 3 discusses how the notional opponent idea makes it possible to produce individual ratings for doubles tennis players. We then discuss possible skill aggregation functions before evaluating them via simulation.

Chapter 4 details the decomposition of a notional opponent into a notional player and notional group. We then show the possible notional opponent mean and variance parameters that are used in the hierarchical Glicko-2 algorithm for doubles tennis. Finally, we compare the sum and mean skill aggregation function for both the hierarchical Glicko-2 algorithm and the Glicko-2 algorithm on real data supplied by the ITF.

Chapter 5 contains some concluding remarks and possible next steps in improving the hierarchical Glicko-2 algorithm for the WTN.

2. Hierarchical Glicko-2

In this chapter, we provide a detailed outline of the Bayesian model behind the hierarchical Glicko-2 algorithm. We begin by introducing the Glicko-2 algorithm, before discussing the player sparsity problem which necessitates the need for a hierarchical variant. Throughout this chapter we discuss the Bayesian model from the perspective of singles tennis, but it could be applied to any scenario where an object follows a Gaussian distribution with comparisons made to other objects that result in a binary outcome variable.

2.1. Bayesian Model for Glicko-2

The Glicko-2 algorithm is an approximate Bayesian updating algorithm (Glickman, 2001). In the general setting, we compare two objects that have Gaussian prior distributions using a binary outcome. When there are many objects to compare, the posterior distribution becomes intractable. Instead of computing the posterior distribution, the marginal posterior distribution is computed by integrating over the prior distributions. While this method does not take into consideration potentially important information, for example, the results of comparisons involving an objects opponent, it does lead to a tractable set of solutions.

In the next subsections we will detail the assumptions made in order to formulate the model for comparing different players.

2.1.1. Player Skill Process

Suppose we have equal time increments and that at time t , K_{ij} tennis matches take place between player i and player j . We do not know the scalar value that represents a player's skill, but this value can be compared with other player's skills to compute the win probability for either player involved in a match. Therefore, we make some distributional assumptions about these unknown values. For each player, i , we assume

that their skill evolves according to the following process

$$\gamma_i^{(t)} | \gamma_i^{(t-1)}, \sigma_i^{2(t)} \sim \mathcal{N}(\gamma_i^{(t-1)}, \sigma_i^{2(t)}), \quad (2.1)$$

$$\log \sigma_i^{2(t)} | \log \sigma_i^{2(t-1)}, \tau^2 \sim \mathcal{N}(\log \sigma_i^{2(t-1)}, \tau^2), \quad (2.2)$$

where τ^2 is constant over time and the same value for all players. It is further assumed to ensure identifiability that $\gamma_i^{(0)} | \omega^2 \sim \mathcal{N}(0, \omega^2)$, with a proper prior density for ω^2 . This type of model allows a player's skill at tennis to change in value and also exhibit sudden changes in variability. These are realistic assumptions since player abilities improve or decline over time, and temporarily unexpected results may be caused by recent events such as returning from injury.

2.1.2. Win Probability Model

With the player skill process in place, we need to consider the probability that player i defeats player j in a tennis match since this will be incorporated in the likelihood term for the calculation of the marginal posterior distribution. The Glicko-2 algorithm was developed by assuming that player win probabilities followed the Bradley-Terry model (Bradley and Terry, 1952). Let $Y_{ijk}^{(t)}$ be the outcome of the k^{th} match between player i and player j at time t . Then $Y_{ijk}^{(t)} = 1$ when player i defeats player j and is 0 otherwise. Since draws are not possible in tennis, we do not consider it here. The Glicko-2 algorithm can be adapted to include draws, but this is not covered in this report. It is also worth pointing out that we could let Y represent the result of a set, the result of a game or even the result of a point. In fact, the current version of the algorithm for the WTN uses sets instead of matches to include more information about the relative abilities of the players involved. Now the probability that player i wins is given by the Bradley-Terry model,

$$P(Y_{ijk}^{(t)} = 1) = \frac{\exp(\gamma_i^{(t)})}{\exp(\gamma_i^{(t)}) + \exp(\gamma_j^{(t)})}.$$

Another probability function could have been used, provided that it is monotonically increasing in $\gamma_i^{(t)} - \gamma_j^{(t)}$. One could even use a probability function that includes covariate information about player i and player j , for example one may conclude that a player's ability is affected by the surface of the court the match takes place on and wish to include this information in the win probability.

2.1.3. Player Posterior Skill

If we assume player i plays M opponents during time t , and let $\mathbf{s}$ represent the outcomes of all the games involving player i , then the marginal posterior distribution of their skill

is,

$$f(\gamma_i^{(t)}|\mathbf{s}) \propto \varphi(\gamma_i^{(t)}|\gamma_i^{(t-1)}, \sigma_i^{2(t)}) \times \prod_{j=1}^{M^{(t)}} \int \prod_{k=1}^{K_{ij}^{(t)}} \frac{(\exp(\gamma_i^{(t)}))^{s_{jk}^{(t)}}}{\exp(\gamma_i^{(t)}) + \exp(\gamma_j^{(t)})} \varphi(\gamma_j^{(t)}|\gamma_j^{(t-1)}, \sigma_j^{2(t)}) d\gamma_j^{(t)} \quad (2.3)$$

where $\varphi(\cdot)$ is the standard Gaussian probability distribution function. This expression could be evaluated using numerical methods, but the Glicko-2 algorithm instead determines a set of closed form computations that approximate the marginal posterior density in equation (2.3). The Glicko-2 algorithm results in a marginal posterior distribution that is Gaussian, $f(\gamma_i^{(t)}|\mathbf{s}) \sim \mathcal{N}(\mu_i^{(t)}, \phi_i^{2(t)})$. The derivation of the mean and variance parameter can be found in Glickman (1999), for now we simply state their formulas,

$$\mu_i^{(t)} = \mu_i^{(t-1)} + \frac{1}{\frac{1}{\phi_i^{2(t)}} + \frac{1}{\nu^2}} \sum_{j=1}^{M^{(t)}} \sum_{k=1}^{K_{ij}^{(t)}} g(\phi_j^{2(t-1)})(s_{jk}^{(t)} - E(s|\mu_i^{(t-1)}, \mu_j^{(t-1)}, \phi_j^{2(t-1)})),$$

$$\phi_i^{2(t)} = \left(\frac{1}{\phi_i^{2(t)}} + \frac{1}{\nu^2} \right)^{-1},$$

where,

$$g(\phi^2) = \frac{1}{\sqrt{1 + \frac{3\phi^2}{\pi^2}}},$$

$$E(s|\mu, \mu_j, \phi_j^2) = \frac{1}{1 + \exp(-g(\phi_j^2)(\mu - \mu_j))},$$

$$\phi_i^{2(t)} = \phi_i^{2(t-1)} + \sigma^{2(t)}.$$

Note that $\sigma^{2(t)}$ needs to be estimated. This is done using the Illinois algorithm (Dowell and Jarratt, 1971), with the exact derivation of $\sigma^{2(t)}$ found in Glickman (2001). Finally, we also have that,

$$\frac{1}{\nu^2} \approx \sum_{j=1}^{M^{(t)}} K_{ij}^{(t)} g(\phi_j^{2(t-1)}) E(s|\mu_i^{(t-1)}, \mu_j^{(t-1)}, \phi_j^{2(t-1)}) (1 - E(s|\mu_i^{(t-1)}, \mu_j^{(t-1)}, \phi_j^{2(t-1)}))$$

Reviewing the above formulae, we highlight that we only require the mean and variance parameters of the prior distributions, as well as the match outcome, to compute the marginal posterior distribution.

The marginal posterior distribution derived via the Glicko-2 updates is then shifted by 1500 and scaled by 173.7178 to produce the player rating, $r_i^{(t)}$, and rating deviation, $RD_i^{(t)}$,

$$r_i^{(t)} = 1500 + 173.7178 \times \mu_i^{(t)},$$

$$RD_i^{(t)} = 173.7178 \times \phi_i^{(t)}.$$

The choice of 1500 and 173.7178 originate from the Elo rating system developed for the United States Chess Federation (USCF) (Elo, 1986). The USCF aimed for the average rating of a player to be 1500, while players with a rating difference of $n \times 400$ over their opponent would have an expected score that is 10^n times their opponents. In the Glicko-2 algorithm, at time $t + 1$ we use the marginal posterior distributions as the prior distributions. In the next section we detail the problems that the Glicko-2 algorithm encounters when applied to tennis data.

2.2. The Player Sparsity Problem

Comparison models, like the one described in the previous section, require many comparisons relative to the number of objects to be able to produce an accurate ranking. We need a large number of comparisons because there is some randomness in the comparisons, in other words if two objects are compared numerous times then it is likely that one object will be preferred only a fraction of the time rather than all the time. However, not only are a large number of comparisons required, but the ‘right’ comparisons need to be made. If two objects have been compared previously then we have some evidence to rank them. If two objects have a common object that they have been compared to then this is also evidence to rank them. A problem occurs when there is a lack of connections between two objects. The ranking of two objects may be reliant upon a string of results between multiple objects. In the data sets collected by the ITF, this situation was found to be a common occurrence. We call this the player sparsity problem, since there are effectively not enough matches played between players to make an accurate ranking.

Player sparsity can allow some players to be misaligned along the single scale. This is most obvious when we look at the highest ranked players by the Glicko-2 algorithm. One would expect the highest ranked players to be professional and making the latter stages of the Grand Slams. If a junior or veteran amateur player manages to make their way in to the top 100 ranked players, then this is more than likely wrong. With the Glicko-2 algorithm unable or slow to accurately align tennis players due to the player sparsity problem, a variant of it was introduced by researchers at Imperial College London called the hierarchical Glicko-2. By decomposing a player’s skill into their individual skill and the skill of the group they play in, we can assign ratings to both players and groups. A player’s individual rating is updated via the Glicko-2 algorithm using all the matches they compete in. When competing against players within their own group we only need to use their individual rating, when competing against players from other groups then we adjust for the groups by using the combined ratings, i.e. the aggregation of the individual and group contributions. The group rating is updated by the Glicko-2 algorithm using all the matches that involve players from that group competing in matches with other groups. In this case the combined ratings are used. The Glicko-2 algorithm can align a player within their group using the results of all the matches they’re involved in, and then align the groups separately. The players can then be ranked on a single scale

using their combined rating. In the next section we explain how the Bayesian model for Glicko-2 can be adapted to include groups.

2.3. Adapting the Bayesian Model to Include Groups

Suppose that the population of tennis players can be grouped into G groups at time t based on the matches played. Within each group there will be many matches played between the players, whereas a smaller percentage of matches played will involve players from different groups. Let player i be assigned to group g and player j assigned to group h . The combined skill of player i , denoted $\gamma_i^{c(t)}$, and player j , denoted $\gamma_j^{c(t)}$, at time t is given by

$$\begin{aligned}\gamma_i^{c(t)} &= \gamma_i^{(t)} + \gamma_g^{(t)}, \\ \gamma_j^{c(t)} &= \gamma_j^{(t)} + \gamma_h^{(t)},\end{aligned}$$

where $\gamma_i^{(t)}$ and $\gamma_j^{(t)}$ are the respective individual contributions and $\gamma_g^{(t)}$ and $\gamma_h^{(t)}$ are the respective group contributions. We assume that both the individual and group contributions evolve as in equations (2.1) and (2.2).

When computing the marginal posterior distribution for player i , we integrate over the prior distributions of player j , group g and group h . The likelihood term contains a linear combination of these variables and since they are all Gaussian deviates, we can combine them in to one Gaussian deviate, denoted $\gamma_{\text{opp}}^{(t)}$ where ‘opp’ stands for opponent. Thus,

$$\gamma_{\text{opp}}^{(t)} = \gamma_j^{c(t)} - \gamma_g^{(t)}. \quad (2.4)$$

By rearranging the Bradley-Terry probability and substituting in $\gamma_{\text{opp}}^{(t)}$, the marginal posterior distribution from equation (2.3) can be expressed as,

$$\begin{aligned}f(\gamma_i^{(t)} | \mathbf{s}) &\propto \varphi(\gamma_i^{(t)} | \gamma_i^{(t-1)}, \sigma_i^{2(t)}) \\ &\times \prod_{j=1}^{M_i^{(t)}} \int \prod_{k=1}^{K_{ij}^{(t)}} \frac{(\exp(\gamma_i^{(t)} - \gamma_{\text{opp}}^{(t)}))^{s_{jk}^{(t)}}}{1 + \exp(\gamma_i^{(t)} - \gamma_{\text{opp}}^{(t)})} \varphi(\gamma_{\text{opp}}^{(t)} | \gamma_{\text{opp}}^{(t-1)}, \sigma_{\text{opp}}^{2(t)}) d\gamma_{\text{opp}}^{(t)}.\end{aligned} \quad (2.5)$$

The Glicko-2 algorithm can now be used to compute the marginal posterior distribution for player i . When player i and player j belong to the same group, we can see that the variables representing the groups will cancel each other out in the likelihood term. In this case, the player skills do not need to be adjusted for groups and the variability is reduced.

Next, we go about updating the marginal posterior distribution for the groups. Consider the model from the perspective of group g . Assume that P players belong to group g .

Let player i belong to group g and player j belong to group h . When the two players in a match are from the same group then the cancellation in the likelihood term means there is no group skill in the integral, so these matches are absorbed into the constant of proportionality. This is useful as we cannot learn about the skill of a group relative to other groups in these matches. The main difference between the Bayesian model for the player skill and the groups skill is that we can integrate with respect to the marginal posterior distributions of the players instead of their prior distributions at time t when computing the marginal posterior for the groups. Thus $\gamma_{\text{opp}}^{(t)}$ will be a linear combination of the marginal posterior distribution of player j and player i as well as the prior distribution of group h . The Glicko-2 algorithm is then applied to the marginal posterior distribution of g ,

$$f(\gamma_g^{(t)}|\mathbf{s}) \propto \varphi(\gamma_g^{(t)}|\gamma_g^{(t-1)}, \sigma_g^2(t)) \times \prod_{i=1}^{P_g^{(t)}} \prod_{j=1}^{M_i^{(t)}} \int \prod_{k=1}^{K_{ij}^{(t)}} \frac{(\exp(\gamma_g^{(t)} - \gamma_{\text{opp}}^{(t)}))^{s_{jk}}}{1 + \exp(\gamma_g^{(t)} - \gamma_{\text{opp}}^{(t)})} \varphi(\gamma_{\text{opp}}^{(t)}|\gamma_{\text{opp}}^{(t-1)}, \sigma_{\text{opp}}^2(t)) d\gamma_{\text{opp}}^{(t)}, \quad (2.6)$$

where $\gamma_{\text{opp}}^{(t)} = \gamma_j^{c(t)} - \gamma_i^{(t)}$. The marginal posterior distribution for the groups will also be Gaussian and we scale it by 173.7178. We do not need to shift the posterior distribution by 1500 since combining the group rating with the player rating should maintain the player average rating at 1500.

2.4. Weaknesses of Using Groups

We raise two potential pitfalls of the hierarchical Glicko-2 algorithm.

Firstly, the skills of the players and the groups are fixed in the actual model, but the approximations made in the Glicko-2 algorithm let the skills be sampled independently from their respective distributions. This can be justified for the skills of players since their level of performance can vary from match to match. However, the skill of a group may be constant at fixed times. Allowing the group skills to be sampled independently increases the uncertainty of the match outcomes, but without the approximation the marginal posterior distribution would be intractable.

Secondly, the matches involving players from different groups will be used to align the relative skills of the groups. However, this assumes that the players that compete against players from other groups are uniformly sampled from their groups. In practice, a selection bias may exist. For example, it's likely that a clustering algorithm will group professional players with each other and junior players with each other based on matches played. It is also more likely that matches between players from the two groups will feature the weakest professionals and strongest juniors. This will cause the hierarchical Glicko-2 algorithm to align the two groups too closely together in terms of average rating.

3. Notional Opponent

In this chapter we discuss the concept of a notional opponent and how it can be used to distil a player’s individual rating from a doubles tennis match. We then describe possible skill aggregation functions that could be used before performing some simulations to determine their suitability. Throughout this chapter we suppress the notation for time, t , and refer to a player’s skill, γ , as a Gaussian distributed random variable with mean μ and variance ϕ^2 to avoid any confusion,

$$\gamma \sim \mathcal{N}(\mu, \phi^2).$$

3.1. Ranking Doubles Tennis Players Individually

In most comparison models two objects are compared with each other, for example a team compared to another team, or a player compared to another player. In doubles tennis the obvious comparison to be made is one pair with their opposing pair in a match. However, we would be left with a ranking system that only contains pairs of players rather than each individual player. This would not be a viable ranking system because each player may be present in multiple rankings if they play with different partners. It is in fact very common to switch partners frequently in doubles tennis. The rating of a partnership may become outdated if they do not play with each other over a period of time. The players may be playing doubles tennis with other partners and improving their skill, but their rating would not reflect this. This represents a loss of important information. When they eventually do revive their partnership, the model will not have an accurate rating of their combined skill. This makes ranking each individual player a much more practical solution.

O’Brien (2019) proposed an adaption to the Bayesian model discussed in the previous chapter that could measure the contribution of each player in a match by adjusting the difficulty of their opponent based on their partner. The term ‘notional opponent’ was coined to describe the opposition pairing adjusted for a players partner. By comparing a player against their notional opponent, we can apply the Bayesian model to doubles tennis. To explain the formulation, consider the following setting: let player 1 be paired with player 2 in a match against player 3 and player 4. The skill of the notional opponent for player 1 is

$$\gamma_{\text{opp}} = f(\gamma_3, \gamma_4) + \gamma_1 - f(\gamma_1, \gamma_2), \quad (3.1)$$

where $f(\cdot, \cdot)$ is a skill aggregation function. The skill aggregation function determines how we measure the combined skill of a pair of tennis players. This formulation allows the players in each pair to have the same expected win probability when compared with their respective notional opponents. If the combined skill of the opposition pair increases then so does the skill of the notional opponent, whereas if the skill of a player's partner increases then the skill of their notional opponent decreases. When the notional opponent is inserted into the Bradley-Terry model, the skill of player 1 cancels out so we are left with,

$$\frac{e^{f(\gamma_1, \gamma_2) - f(\gamma_3, \gamma_4)}}{1 + e^{f(\gamma_1, \gamma_2) - f(\gamma_3, \gamma_4)}}. \quad (3.2)$$

There is one issue with the formulation of the notional opponent derived in O'Brien (2019). It allows a player's own skill to be incorporated in the skill of their notional opponent. This violates the assumptions made in the Glicko-2 algorithm since a player's skill is held constant and the skill of their opponent is integrated out. We propose using the win probability model in equation (3.2) and allowing the skill of the notional opponent to incorporate the skill of all players involved except the player we are updating. The choice of skill aggregation function also means we may not end up comparing a player's skill against a notional opponent. For example, if we use the mean of a pair's skill then we will be comparing half of a player's skill against their notional opponent. Fortunately, the Glicko-2 algorithm can be adapted to deal with this. In the next section we propose some potential skill aggregation functions.

3.2. Skill Aggregation Functions

During this project we examined 4 possible ways determine the skill of a pair of tennis players. We used the sum, mean, maximum, and minimum of their individual skills. In the following subsections we derive the distribution of the pair's skill for the respective aggregation functions. Each player's skill follows a Gaussian distribution, so any linear combination of the player's skills will result in a Gaussian distribution. Using the maximum or minimum function will not produce a Gaussian distribution, but the Glicko-2 algorithm only requires a mean and a variance parameter. This means the Glicko-2 algorithm can still be applied to functions of players skills that do not produce a Gaussian distribution, but we expect the results to be negatively affected by this.

The choice of skill aggregation function to examine are subjective and the list we have studied is not exhaustive. Attempts to add or average player ability seem like appropriate methods that do not need justification. The maximum and minimum, however, puts more emphasis on an individual's skill determining the outcome of the match. They are inspired by the idea that a team is only as fast as its slowest member, or in this case a tennis pairing is only as good as the stronger, or alternatively weaker, player on match day.

In the next subsections we derive the distribution of the notional opponent for player 1 when they are partnered with player 2 and facing player 3 and player 4.

3.2.1. Sum

The sum aggregation function is relatively straightforward and defined as

$$f(\gamma_i, \gamma_j) = \gamma_i + \gamma_j.$$

By adding the skill of two teammates and subtracting one of the opposition player's skill, we produce a Gaussian distributed random variable. Therefore, the skill of player 1's notional opponent is given by

$$\gamma_{\text{opp}} \sim \mathcal{N}(\mu_3 + \mu_4 - \mu_2, \phi_3^2 + \phi_4^2 + \phi_2^2).$$

3.2.2. Mean

The mean aggregation function is also simple to define,

$$f(\gamma_i, \gamma_j) = \frac{\gamma_i + \gamma_j}{2}.$$

The mean also produces a Gaussian distributed random variable for the skill of player 1's notional opponent,

$$\gamma_{\text{opp}} \sim \mathcal{N}\left(\frac{\mu_3 + \mu_4 - \mu_2}{2}, \frac{\phi_3^2 + \phi_4^2 + \phi_2^2}{4}\right).$$

However, in this case we do not have player 1's full skill in the model. Instead we have half of player 1's skill, so the Glicko-2 algorithm is adapted to show this via the formulae for the posterior mean and variance parameters,

$$\begin{aligned} \mu_i^{(t)} &= \mu_i^{(t-1)} + \frac{1}{\frac{1}{\phi_i^{2(t)}} + \left(\frac{1}{2}\right)^2} \sum_{j=1}^{M^{(t)}} \sum_{k=1}^{K_{ij}^{(t)}} \frac{g(\phi_j^{2(t-1)})}{2} (s_{jk}^{(t)} - E(s | \frac{\mu_i^{(t-1)}}{2}, \mu_j^{(t-1)}, \phi_j^{2(t-1)})), \\ \phi_i^{2(t)} &= \left(\frac{1}{\phi_i^{2(t)}} + \left(\frac{1}{2}\right)^2 \right)^{-1}. \end{aligned}$$

The derivation can be found in Appendix A.

3.2.3. Maximum

The maximum skill aggregation function is defined as

$$f(\gamma_i, \gamma_j) = \max(\gamma_i, \gamma_j).$$

The maximum does not produce a Gaussian distribution for the notional opponent's skill, but we can use the mean and variance parameters for the Glicko-2 algorithm,

$$\begin{aligned}\mu_{\text{opp}} &= \mathbb{E}[\max(\gamma_3, \gamma_4)] + \mu_1 - \mathbb{E}[\max(\gamma_1, \gamma_2)], \\ \phi_{\text{opp}}^2 &= \text{Var}[\max(\gamma_3, \gamma_4)] + \phi_1^2 + \text{Var}[\max(\gamma_1, \gamma_2)] - 2 \text{Cov}(\gamma_1, \max(\gamma_1, \gamma_2)).\end{aligned}$$

To compute the mean and variance of the notional opponent we can use the first and second moments of $\max(\gamma_i, \gamma_j)$. The moments are given by equation (3.3) and equation (3.4), while the derivation can be found in Nadarajah and Kotz (2008),

$$\begin{aligned}\mathbb{E}[\max(\gamma_i, \gamma_j)] &= \mu_i \Phi\left(\frac{\mu_i - \mu_j}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right) + \mu_j \Phi\left(\frac{\mu_j - \mu_i}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right) \\ &\quad + (\phi_i^2 + \phi_j^2)^{\frac{1}{2}} \varphi\left(\frac{\mu_i - \mu_j}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right),\end{aligned}\tag{3.3}$$

$$\begin{aligned}\mathbb{E}[\max(\gamma_i, \gamma_j)^2] &= (\phi_i^2 + \mu_i^2) \Phi\left(\frac{\mu_i - \mu_j}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right) + (\phi_j^2 + \mu_j^2) \Phi\left(\frac{\mu_j - \mu_i}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right) \\ &\quad + (\mu_i + \mu_j)(\phi_i^2 + \phi_j^2)^{\frac{1}{2}} \varphi\left(\frac{\mu_i - \mu_j}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right).\end{aligned}\tag{3.4}$$

The standard Gaussian cumulative distribution function is denoted by $\Phi(\cdot)$ in equation (3.3) and equation (3.4). Lastly, the covariance term is given by equation (3.5) and the derivation is given in Appendix A,

$$\text{Cov}(\gamma_i, \max(\gamma_i, \gamma_j)) = \phi_i^2 \Phi\left(\frac{\mu_i - \mu_j}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right).\tag{3.5}$$

3.2.4. Minimum

The results for the minimum skill aggregation function are very similar to those of the maximum skill aggregation function. The minimum skill aggregation function is defined as

$$f(\gamma_i, \gamma_j) = \min(\gamma_i, \gamma_j).$$

The mean and variance parameters for notional opponent's skill are

$$\begin{aligned}\mu_{\text{opp}} &= \mathbb{E}[\min(\gamma_3, \gamma_4)] + \mu_1 - \mathbb{E}[\min(\gamma_1, \gamma_2)], \\ \phi_{\text{opp}}^2 &= \text{Var}[\min(\gamma_3, \gamma_4)] + \phi_1^2 + \text{Var}[\min(\gamma_1, \gamma_2)] - 2 \text{Cov}(\gamma_1, \min(\gamma_1, \gamma_2)).\end{aligned}$$

To compute these parameters we require the first and second moments of $\min(\gamma_i, \gamma_j)$ which are given by equation (3.6) and equation (3.7), with the derivation again found in Nadarajah and Kotz (2008),

$$\begin{aligned} \mathbb{E}[\min(\gamma_i, \gamma_j)] &= \mu_i \Phi\left(\frac{\mu_j - \mu_i}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right) + \mu_j \Phi\left(\frac{\mu_i - \mu_j}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right) \\ &\quad - (\phi_i^2 + \phi_j^2)^{\frac{1}{2}} \varphi\left(\frac{\mu_j - \mu_i}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right), \end{aligned} \quad (3.6)$$

$$\begin{aligned} \mathbb{E}[\min(\gamma_i, \gamma_j)^2] &= (\phi_i^2 + \mu_i^2) \Phi\left(\frac{\mu_j - \mu_i}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right) + (\phi_j^2 + \mu_j^2) \Phi\left(\frac{\mu_i - \mu_j}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right) \\ &\quad - (\mu_i + \mu_j)(\phi_i^2 + \phi_j^2)^{\frac{1}{2}} \varphi\left(\frac{\mu_i - \mu_j}{(\phi_j^2 + \phi_i^2)^{\frac{1}{2}}}\right). \end{aligned} \quad (3.7)$$

Finally, the covariance term is given by equation (3.8) and the derivation is given in Appendix A,

$$\text{Cov}(\gamma_i, \max(\gamma_i, \gamma_j)) = \phi_i^2 \Phi\left(\frac{\mu_j - \mu_i}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right). \quad (3.8)$$

3.3. Notional Opponent Simulations

To evaluate the skill aggregation functions we can make use of fictitious data. By generating data that we know the truth values to, e.g. the win probabilities, the player ratings and rating deviations, we can begin to understand the strengths and weaknesses of each skill aggregation function. Since we are not including groups in our simulation, we are using the Glicko-2 algorithm and not the hierarchical variant. The following subsections discuss how we generated the player population, the matches, the match results and finally, the performance of the skill aggregation functions.

3.3.1. Player Generation

Currently, there are 1,482 professional female tennis players and 2,137 male tennis players (ITF, 2021). Professional players will make up the largest group of players the Glicko-2 algorithm will likely have to deal with, so in our simulation we generated 2,000 players. To generate a player, we first initialise their rating ($r^{(0)}$), rating deviation ($RD^{(0)}$) and rating volatility (σ). The volatility parameter is fixed, while the rating and rating deviation vary with time. We allowed the mean on each players prior rating to be sampled from a normal distribution centred at 1500 with a much larger variance relative to the square of the rating deviation. It is more realistic that player ratings differ greatly within a group but do not evolve with large changes. At time $t = 0$, for each player, i ,

we sample

$$\begin{aligned}\log(\sigma_i) &\sim \mathcal{N}(\log(0.06), 0.001^2), \\ \log(RD_i^{(0)}) &\sim \mathcal{N}(\log(10), \sigma_i^2), \\ \mu_i &\sim \mathcal{N}(1500, 350), \\ r_i^{(0)} &\sim \mathcal{N}(\mu_i, RD_i^{2(0)}).\end{aligned}$$

Thus, each player has an initial skill that is Gaussian distributed with mean $r^{(0)}$ and standard deviation $RD^{(0)}$. Deciding what are appropriate priors depends on the setting. It is worth considering how much a player's expected win probability should change from month to month against an opponent with the same initial rating but held constant. For example, if a players rating improves from 1500 to 1800 in 1 month, then their expected win probability jumps from 50% to around 85%. Such a sudden change in a short space of time is unlikely in tennis. At time $t + 1$ each player's skill evolves as follows,

$$\begin{aligned}\log(RD_i^{(t+1)}) &\sim \mathcal{N}(\log(RD_i^{(t)}), \sigma_i^2), \\ r_i^{(t+1)} &\sim \mathcal{N}(r_i^{(t)}, RD_i^{2(t+1)}).\end{aligned}$$

3.3.2. Match Generation

The current implementation of the hierarchical Glicko-2 algorithm for the WTN uses up to 4 years of tennis match data at the set level to determine a player's rating and rating deviation, which is then used to assign their WTN. For this reason, we generated matches over a 4-year period with monthly rating periods, so there were 48 rating periods in total starting at $t = 0$. Glickman (1999) recommends between 5 and 10 matches for each player in each rating period to give some indication of player skill for the Glicko-2 algorithm. In each rating period we generated 5,000 matches since this came to an average of 10 games per player. In total 240,000 matches were generated over 4 years. To choose each player in each game we followed this iterative procedure,

1. Select Player 1:

The initial player involved in the match is chosen uniformly at random from the entire population of players.

2. Select Player 1's teammate:

If Player 1 has played previously, then 90% of the time we let their teammate be the player they previously played with. The other 10% of the time their teammate

will be chosen based on the player ratings by sampling from a Gaussian density centred at Player 1's rating with the standard deviation equal to the average of the maximum and minimum player ratings in the population, with the probabilities then normalised. This will also be used for when Player 1 has not previously played.

3. Select Opposition Player 1:

Opposition Player 1 will be chosen by sampling from a Gaussian density centred at Player 1's rating with the standard deviation equal to the average of the maximum and minimum player ratings in the population, with the probabilities then normalised.

4. Select Opposition Player 1's teammate:

Opposition Player 1's teammate will be chosen using the same process as Player 1's teammate but the Gaussian density will be centred at Opposition Player 1's rating.

The selection of player teammates and the opposition in the above procedure was designed to mimic realistic scenarios. Players will tend to play multiple matches in a row with the same teammate, e.g. in a knockout tournament. Players also tend to play against opposition that are of a similar level, e.g. professionals compete in the same tournaments.

3.3.3. Match Simulation

Each match is assumed to be independent with a binary outcome, i.e. either pair 1 wins or pair 2 wins. Thus the result of each match can be simulated using a Bernoulli distribution, where the probability that pair 1 wins is calculated via the Bradley-Terry model in equation (3.2). This model requires as input the combined skill of each pair using the skill aggregation function. Since each aggregation function produces different ratings, the match results will differ not just through randomness from the Bernoulli sampling but also from the win probabilities. For each aggregation function, we need to assume that they are the correct underlying function when simulating match results. Without this assumption there is no link between match results and player skill. As a consequence, the aggregation functions are not comparable, but the aim of this simulation is to develop an understanding of how each one behaves.

3.3.4. Simulation Results

To learn about the skill aggregation functions we ran the Glicko-2 algorithm using each function on their corresponding simulated data set. At the end of the 4 years we evaluated the performance of each aggregation function. For the sum aggregation function, Figure 3.1 shows a scatterplot comparing the actual player ratings at the final rating period against the estimated player rating via the Glicko-2 algorithm using the sum aggregation function. The actual 95% confidence interval for each player’s rating runs horizontally through each point while the estimated 95% credible interval for each player’s rating runs vertically through each point. We note the linear relationship between the actual player ratings and the estimated player ratings. This shows the algorithm is correctly rating players on average. Figure 3.2 displays the scatterplot of actual player ratings against the estimated player ratings for the mean aggregation function. Once again, we see the desired linear relationship. The maximum and minimum aggregation functions fail to accurately rate the players ratings, per Figure 3.3 and Figure 3.4 respectively. The weakest players are rated poorly for the maximum, while the strongest players are rated poorly for the minimum. This makes intuitive sense since the worst player will never impact the win probability when using the maximum and the best player will never impact the win probability when using the minimum. These player’s match results, and consequently Glicko-2 rating, will be determined by their teammate.

Table 3.1 shows the root mean squared error (RMSE), mean absolute error (MAE) and Spearman rank correlation (Spearman ρ) between the actual ratings and the estimated ratings for all the skill aggregation functions. The sum aggregation function produces the minimum RMSE and MAE, as well as the maximum Spearman ρ between all the skill aggregation functions. This leads to more accurate ratings and rankings. However, the data sets are not the same for each aggregation function so they cannot be compared directly in this way. For example, the win probabilities for a pair that is favourite in a match (i.e. 50% chance or more) computed using the sum aggregation function will always be larger than the win probabilities computed using the mean aggregation function. Nonetheless, the figures and statistics are evidence to justify the exclusion of the maximum and minimum skill aggregation functions. The optimal choice of skill aggregation function will ultimately depend on the data set that it is applied to.

Table 3.1.: The root mean squared error (RMSE), mean absolute error (MAE) and Spearman rank correlation (Spearman ρ) between the actual player ratings and the estimated player ratings from each skill aggregation functions simulation at $t = 47$.

Function	RMSE	MAE	Spearman ρ
Sum	52	40	0.99
Mean	85	65	0.98
Maximum	147	96	0.95
Minimum	142	95	0.95

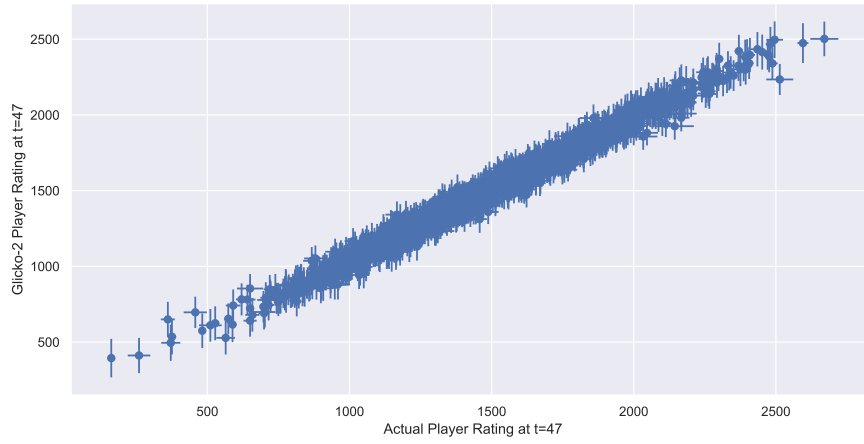


Figure 3.1.: The actual player ratings against the estimated player ratings using the Glicko-2 algorithm with the sum skill aggregation function. Error bars represent the 95% Credible intervals.

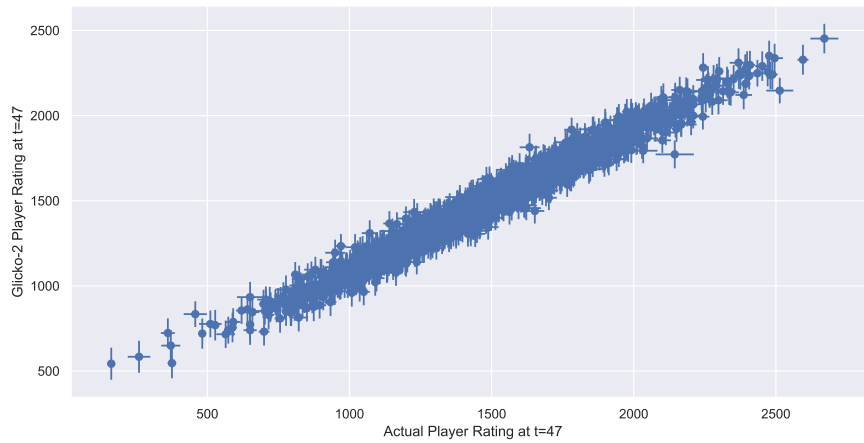


Figure 3.2.: The actual player ratings against the estimated player ratings using the Glicko-2 algorithm with the mean skill aggregation function. Error bars represent the 95% Credible intervals.

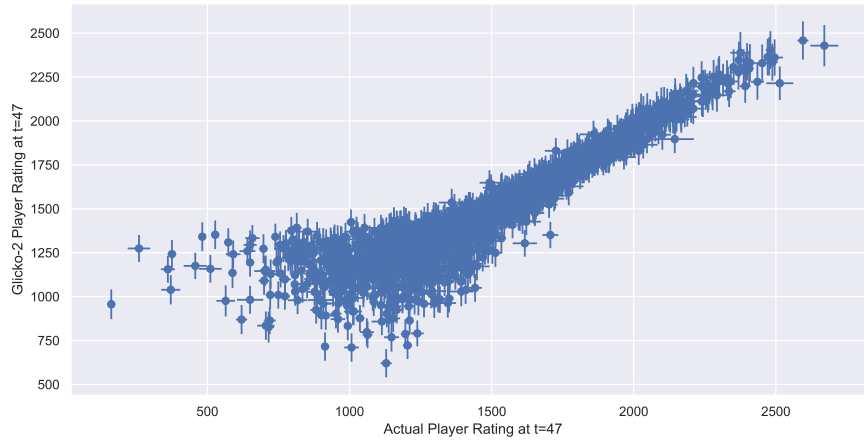


Figure 3.3.: The actual player ratings against the estimated player ratings using the Glicko-2 algorithm with the maximum skill aggregation function. Error bars represent the 95% Credible intervals.

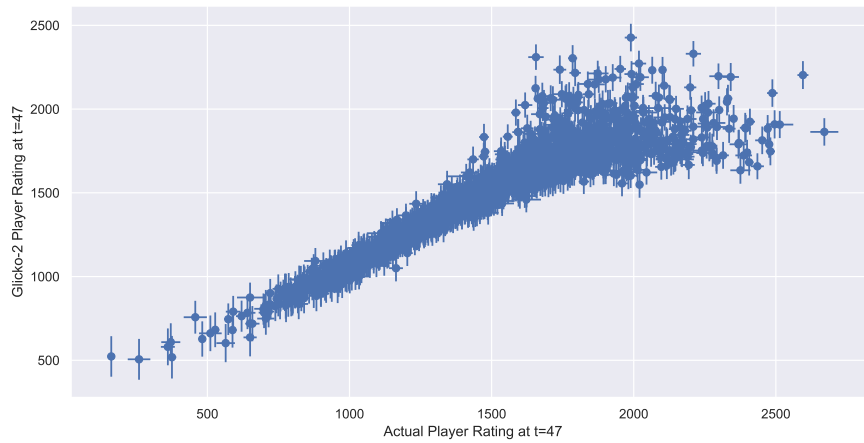


Figure 3.4.: The actual player ratings against the estimated player ratings using the Glicko-2 algorithm with the minimum skill aggregation function. Error bars represent the 95% Credible intervals.

4. Notional Groups

In the previous chapters, we discussed how to adapt the underlying Bayesian model to include groups and pairs of players. In this section, we merge these ideas to create a notional player and a notional group. We derive the possible notional opponents for both players and groups using both the sum aggregation function and the mean aggregation function. Finally we assess these aggregation functions suitability for doubles tennis.

4.1. Aligning Group Skill

Rating groups using the hierarchical Glicko-2 algorithm allows for a faster and more accurate ranking of all players on a single scale. For singles tennis, rating groups is straightforward. For example, whenever a player in group A plays a match against a player in group B, then the result can be used to determine both groups relative ratings. If on average players from group A beat players from group B 70% of the time, then the group ratings will reflect this. Obviously for doubles tennis it is slightly more complicated since there are 4 players involved. We can extend the concept of a notional opponent to the groups.

A notional group incorporates the strength of all the other groups involved in a tennis match. Thus when determining the expected win probability for a player or a group in a doubles tennis match, we can decompose the combined skills of their teammate and opposition players into a player element, the ‘notional player’, and a group element, the ‘notional group’. For doubles tennis with players assigned to different groups, the notional opponent is the aggregation of the notional player and notional group

The main difference between the notional player and the notional group is that the same group may be present multiple times whereas a player can only be present once in the calculation of a notional player. For example, two players on the same team that are in the same group will increase the variability in our win predictions because the group element of their skill is sampled from the same process. There will also be cases where two opposing players are from the same group, resulting in a cancellation of the group elements. In the next section we go through all the possible cases to compute the notional opponent ratings and rating deviations for both players and groups.

4.2. Setup

Consider a doubles tennis match involving four players: player 1, player 2, player 3 and player 4. Let player 1 and player 2 be paired against player 3 and player 4. Now assume that four groups exist: group A, group B, group C and group D. We will assign the players to their respective groups in the case-by-case analysis in the next section. The skill of a player, say i , and group, say I , are defined below,

$$\begin{aligned}\gamma_i &\sim \mathcal{N}(\mu_i, \phi_i^2) \\ \gamma_I &\sim \mathcal{N}(\mu_I, \phi_I^2)\end{aligned}$$

Furthermore, let $\gamma_i^c = \gamma_i + \gamma_I$, be defined as the combined skill for a player. The notional player and notional group skill will all be Gaussian when using the sum or mean aggregation function. In the next sections we derive the mean and variance parameter for each from the perspective of player 1 and from the perspective of group A for both aggregation functions. We will always assign player 1 to group A, without loss of generality.

4.3. Notional Player and Notional Group

In the next subsections we go through the mean and variance parameters for the notional opponent of a player and of a group using the sum aggregation function. In each case we detail the notional player and notional group element. The results for the mean aggregation function are the same as the results for the sum aggregation function, but with μ_{opp} divided by 2 and ϕ_{opp}^2 divided by 4.

4.3.1. Player Opponent

For a player, the notional opponent mean and variance parameters will depend on the group each player belongs to. To simplify the equations we let the notional opponent's skill be decomposed in to the player element and the group element,

$$\gamma_{\text{opp}} = \gamma_p + \gamma_g$$

For player 1, the notional player will be the same regardless of the groups the other players belong to. Therefore the mean and variance parameter of the notional player are,

$$\begin{aligned}\mu_p &= \mu_4 + \mu_3 - \mu_2 \\ \phi_p &= \phi_4^2 + \phi_3^2 + \phi_2^2\end{aligned}\tag{4.1}$$

To compute the notional group we examine 7 possible cases. Each case may contain different scenarios, but ultimately, they arrive at the same notional group.

4.3.2. Group Opponent

The notional opponent for a group also contains a player element that does not change, but it includes player 1's skill so it has different parameters compared to the formulae in (4.1),

$$\begin{aligned}\mu_p &= \mu_4 + \mu_3 - \mu_2 - \mu_1 \\ \phi_p^2 &= \phi_4^2 + \phi_3^2 + \phi_2^2 + \phi_1^2\end{aligned}$$

Recall that the idea of a notional opponent for groups is to pit γ_A versus γ_{opp} . There are some scenarios, which we outline in the following subsections, where we have $2\gamma_A$ versus γ_{opp} . The Glicko-2 algorithm can still be applied as normal, but we need to count the game twice for group A. In most cases we only ever count a game once for a group. The only other special case occurs when there is a cancellation in the group elements, which results in the game not being used to align the groups. This is an important distinction between the algorithm for the groups and algorithm for the players. When updating player ratings, a game is used 4 times in the hierarchical Glicko-2 algorithm (once for each player). When updating group ratings, a game may be used between 4 and 0 times, depending on which groups the players belong to.

Case 1

Notional group parameters for notional opponent of player 1:

$$\begin{aligned}\mu_g &= 0 \\ \phi_g^2 &= 0\end{aligned}$$

There is no notional opponent for group A in this case due to group A's skill being cancelled out in the likelihood term for the Bayesian model.

In the first case, there is no group element in the notional opponent because the groups cancel each other out. This can happen when all 4 players are in the same group, or when player 1 is in the same group as one of the opposition players and their teammate is in the same group as the other opposition player. Below we list the groups each player must belong to for there to be no group element,

Sub-Case	Player 1	Player 2	Player 3	Player 4
1	A	A	A	A
2	A	B	A	B
3	A	B	B	A

Thus these types of matches cannot be used to update the rating of group A. We ignore all these cases for group A, but this does not mean that the matches cannot be used to

update the ratings of the other groups that may be involved. Sub-case 2 is likely to be a commonly occurring scenario in mixed doubles tennis. For example, both pairs may contain a professional male and a professional female tennis player. Aligning groups of male and female tennis players may be difficult if there is a lack of data.

Case 2

Notional group parameters for notional opponent of player 1:

$$\begin{aligned}\mu_g &= \mu_B - \mu_A \\ \phi_g^2 &= \phi_B^2 + \phi_A^2\end{aligned}$$

Notional group parameters for notional opponent of group A:

$$\begin{aligned}\mu_g &= \mu_B \\ \phi_g^2 &= \phi_B^2\end{aligned}$$

There are five sub-cases that result in the mean and variance parameters for the notional opponent of player 1 and of group A for case 2. The first two sub-cases are when player 1 is in the same group as their teammate and also as one of the opposition players. The next two sub-cases are when player 1's teammate is in the same group as one of the opposition players and then the other opposition player is in a different group. Finally, the mean and variance parameters also occur when players 1's teammate and both opposition players are in the same group. These sub-cases may again be best understood in table format,

Sub-Case	Player 1	Player 2	Player 3	Player 4
1	A	A	B	A
2	A	A	A	B
3	A	C	B	C
4	A	C	C	B
5	A	B	B	B

Note that despite group A appearing multiple times in sub-case 1 and sub-case 2, we only count the game once for group A in the hierarchical Glicko-2 algorithm. Thus we only need to update group A using sub-case 1 or sub-case 2, but not both.

Case 3

Notional group parameters for notional opponent of player 1:

$$\begin{aligned}\mu_g &= \mu_A - \mu_B \\ \phi_g^2 &= \phi_A^2 + \phi_B^2\end{aligned}$$

Notional group parameters for notional opponent of group A:

$$\begin{aligned}\mu_g &= -\mu_B \\ \phi_g^2 &= \phi_B^2\end{aligned}$$

Case 3 does not have any sub-cases since it only occurs when player 1 is in the same group as both opposition players, but their teammate is in a different group.

Sub-Case	Player 1	Player 2	Player 3	Player 4
1	A	B	A	A

If we reverse the ordering of the pairs, we see that this case is already dealt with in sub-case 1 of case 2. We do not need to update group A's rating using this match since it can be updated using sub-case 1 of case 2.

Case 4

Notional group parameters for notional opponent of player 1:

$$\begin{aligned}\mu_g &= 2\mu_B - 2\mu_A \\ \phi_g^2 &= 4\phi_A^2 + 4\phi_B^2\end{aligned}$$

Notional group parameters for notional opponent of group A:

$$\begin{aligned}\mu_g &= 2\mu_B \\ \phi_g^2 &= 4\phi_B^2\end{aligned}$$

Case 4 is likely to be commonly occurring in tennis match play since it happens when player 1 is in the same group as their teammate, while the opposition players are also in the same group together. This case results in an increased variation. For group A we have $2\gamma_A$ compared to its notional opponent γ_{opp} . In the hierarchical Glicko-2 algorithm we use this match twice to update group A's rating.

Sub-Case	Player 1	Player 2	Player 3	Player 4
1	A	A	B	B

Case 5

Notional group parameters for notional opponent of player 1:

$$\begin{aligned}\mu_g &= \mu_B + \mu_C - 2\mu_A \\ \phi_g^2 &= \phi_B^2 + \phi_C^2 + 4\phi_A^2\end{aligned}$$

Notional group parameters for notional opponent of group A:

$$\begin{aligned}\mu_g &= \mu_B + \mu_C \\ \phi_g^2 &= \phi_B^2 + \phi_C^2\end{aligned}$$

Case 5 is a slight variation to case 4. It happens when player 1 is in the same group as their teammate but both opposition players are in different groups to each other. There are two sub-cases since we can permute the groups of the opposition players. Like case 4, we have $2\gamma_A$ compared to γ_{opp} so we can use these matches twice to update the rating of group A.

Sub-Case	Player 1	Player 2	Player 3	Player 4
1	A	A	B	C
2	A	A	C	B

Case 6

Notional group parameters for notional opponent of player 1:

$$\begin{aligned}\mu_g &= 2\mu_C - \mu_A - \mu_B \\ \phi_g^2 &= 4\phi_C^2 + \phi_B^2 + \phi_A^2\end{aligned}$$

Notional group parameters for notional opponent of group A:

$$\begin{aligned}\mu_g &= 2\mu_C - \mu_B \\ \phi_g^2 &= 4\phi_C^2 + \phi_B^2\end{aligned}$$

Case 6 occurs when player 2 is in a different group to player 1, while both opposition players are in the same group.

Sub-Case	Player 1	Player 2	Player 3	Player 4
1	A	B	C	C

Case 7

Notional group parameters for notional opponent of player 1:

$$\begin{aligned}\mu_g &= \mu_C - \mu_B \\ \phi_g^2 &= \phi_C^2 + \phi_B^2\end{aligned}$$

There is no notional opponent for group A in this case due to group A's skill being cancelled out in the likelihood term for the Bayesian model.

The final case happens when player 1 is in the same group as one of the opposition players, but their teammate and the other opposition player are in different groups. Since the γ_A is cancelled out, we do not use this type of match to update the rating of group A.

Sub-Case	Player 1	Player 2	Player 3	Player 4
1	A	B	A	C
2	A	B	C	A

4.4. Ranking Tennis Players in ITF Competitions

The ITF organizes doubles competitions and matches every year for junior, veteran, and professional players. This means they have access to tennis results involving all these players. To compare the sum and mean skill aggregation function we used a data set compiled by the ITF. The data set contained tennis set results over 4 years from 2016 to 2020. In total there were 496,075 tennis sets played with 65,216 tennis players recorded as having played at least 1 set.

Using real data does not present us with the true value for a players skill like in the simulations performed previously. Thus we must judge the performance of the skill aggregation functions using some sort of metric. In the next subsections we describe methods for assessing the ratings derived from the hierarchical Glicko-2 algorithm.

4.4.1. Upset Ratio

The upset ratio is defined as the number of sets that the underdog, as predicted by the algorithm, won compared to the total number of sets. The win probability for a player, say i , against their notional opponent is given by,

$$\frac{1}{1 + e^{-g(\phi_i^2 + \phi_{\text{opp}}^2)(\mu_i - \mu_{\text{opp}})}} \quad (4.2)$$

where

$$g(\phi^2) = \frac{1}{\sqrt{1 + \frac{3\phi^2}{\pi^2}}}$$

We use the parameters from the hierarchical Glicko-2 algorithm to compute the win probability for a player. Note that the win probability for a player is the same for their pair. Thus whenever a pair has an expected win probability of less than 0.5 and they win, we record this as a win for the underdog. A model that picks winners and losers of a tennis set by flipping a coin, e.g. heads for pair 1 and tails for pair 2, would be expected to have an upset ratio of 0.5. If the hierarchical Glicko-2 produces accurate

relative ratings of players, then the expected win probabilities will be close to the long run frequencies and the upset ratio should be lower than 0.5.

4.4.2. Log Loss

One weakness of using the upset ratio is that it may be too crude. For example, it does not take into consideration how close the predicted win probability is to the observed win probability. Therefore we also use the log loss to aid our choice of aggregation function. The log loss is defined as,

$$\text{log loss} = -\frac{1}{N} \sum_{i=1}^N y_i \log(p(y_i)) + (1 - y_i) \log(1 - p(y_i))$$

where $y_i = 1$ when player i wins a set and 0 otherwise, and $p(y_i)$ is the expected win probability from equation (4.2). If the observed win probabilities for tennis sets is the same as the expected win probabilities, then the log loss will be small. Thus we seek to minimise the log loss.

4.4.3. Comparison with ATP and WTA Rankings

Lastly, it is possible to produce player ratings that lead to a very low upset ratio and log loss but still be poor when judged subjectively. For example, if a tennis ranking system puts a veteran or junior player above top professionals like Novak Djokovic or Ashleigh Barty then it will never be respected. It is possible for this to happen in the hierarchical Glicko-2 algorithm since Djokovic and Barty may never play veterans or juniors and the groups they are assigned to may not interact. This means that this type of problem would not be identified using just the upset ratio or log loss. Therefore, we compare the rankings from the hierarchical Glicko-2 algorithm with the rankings of professional tennis organisations.

Male and female professional tennis players receive a doubles world ranking from the Association of Tennis Professionals (ATP) and the Women's Tennis Association (WTA), respectively. These world rankings do not include veteran or junior players so cannot be used to judge the performance of the hierarchical Glicko-2 algorithm alone, but it will highlight obvious errors if the best ranked ATP and WTA players are not ranked highly by the algorithm. We do not expect the algorithm's ranking of professional players to match the world rankings provided by these two organisations since they have different methods for determining their ranking, including giving certain tournaments more weighting. We would also expect the top professionals that mainly compete in Singles competition to be ranked higher by the hierarchical Glicko-2 algorithm compared to the ATP or WTA rankings.

To compare the rankings we can use the mean absolute error (MAE) and root mean squared error (RMSE) of our rankings compared to the ATP and WTA rankings for the male and female professionals. We also look at the number of players in our top 500 that are not in the top 500 players ranked by the ATP (for males) and WTA (for females). We aim to minimise all these metrics.

4.4.4. Learning Period and Hyperparameters

To show that the hierarchical Glicko-2 algorithm is an improvement on the Glicko-2 algorithm we evaluated the rankings made by both algorithms using the mean and sum aggregation functions on the data collected by the ITF. While we have 4 years' worth of data, we only assess the results of the algorithms on the final 18 months. The first 30 months are used as a learning period to produce stable initial ratings. We also do not include sets with new players when assessing the performance of the algorithm since we do not know the skill of these players, so we could not make an accurate prediction. A weekly rating period was used for both skill aggregation functions and both algorithms.

To this point we have not discussed how we group players. The hierarchical Glicko-2 algorithm clusters players using the Leiden algorithm developed by Traag et al. (2019). This algorithm requires a resolution parameter and a beta parameter. We used a grid search to identify the hyperparameters that produced the best grouping, by minimising the number of players that were in the top 500 of the hierarchical Glicko-2 rankings but not in the top 500 of the WTA or ATP rankings. We found setting the Leiden resolution to 0.8 and the Leiden beta to 0.07 produced the best results for both the mean and sum aggregation functions. Finally, we also allowed the number of iterations to be 3. By setting the Leiden resolution to 0, the Leiden algorithm will produce only one group. This can be used to run the Glicko-2 algorithm. A more detailed overview of the Leiden algorithm used for the WTN can be found in Depasse (2020) and Cauchetier (2021). The code used for the hierarchical Glicko-2 algorithm is on a private GitHub page and for confidentiality reasons cannot be accessed without approval.

4.4.5. Results

Figure 4.1 shows the upset ratio during each rating period for the hierarchical Glicko-2 algorithm and the Glicko-2 algorithm using both the mean aggregation function and the sum aggregation function. The upset ratios tend to vary around the same values. The hierarchical Glicko-2 algorithm with the mean aggregation function produced the lowest upset ratio 39% of the time. Table 4.1 shows the total upset ratio over the entire period. The hierarchical Glicko-2 algorithm with the mean aggregation function had the lowest upset ratio by 0.0021, which represented 138 less upsets than the Glicko-2 algorithm with the mean aggregation function.

The log loss can give us a more informative view of the predictions made using each function. Per Figure 4.2, the mean aggregation function clearly performs better than the sum aggregation function for both the hierarchical Glicko-2 algorithm and the Glicko-2 algorithm. Over the total period the log loss of the Glicko-2 algorithm with the mean aggregation function was 0.0009 less than the log loss of the hierarchical Glicko-2 algorithm with the mean aggregation function.

Figure 4.3 shows the expected win probabilities against the observed win probabilities for the hierarchical Glicko-2 algorithm with the sum aggregation function, while the same plot is shown for the hierarchical Glicko-2 algorithm with the mean aggregation function in Figure 4.4, the Glicko-2 algorithm with the sum aggregation function in Figure 4.5, and lastly the Glicko-2 algorithm with the mean aggregation function in Figure 4.6. Using the sum aggregation function, the algorithm predicted underdogs won more often than initially predicted, while the opposite is true for the favourites since this is a binary outcome. As an example, on average pairs that were predicted to win 10% of the time actually won just over 20% of the time for the hierarchical Glicko-2 algorithm with the sum aggregation function. The expected win probability using the mean aggregation function was much closer to the observed for both the Glicko-2 algorithm and the hierarchical Glicko-2 algorithm. However, the Glicko-2 algorithm with the mean aggregation function appears to struggle to accurately predict sets with heavy favourites. For example, players that were expected to win 95% of the time actually won over 99% of the time.

Table 4.1.: Total upset ratio and total log loss for the hierarchical Glicko-2 algorithm and the Glicko-2 algorithm using the sum and mean skill aggregation functions based on 18 months of ITF tennis set data.

Function	Total Upset Ratio	Total Log Loss
Sum (hierarchical Glicko-2)	0.3501	0.6240
Mean (hierarchical Glicko-2)	0.3485	0.6111
Sum (Glicko-2)	0.3506	0.6198
Mean (Glicko-2)	0.3492	0.6102

While the upset ratio and log loss showed that the mean aggregation function was superior to the sum aggregation function for both algorithms, the results were split when comparing the mean aggregation function for the hierarchical Glicko-2 algorithm and for the Glicko-2 algorithm. Furthermore, we have yet to analyse the rankings subjectively. Table 4.2 shows the statistics used to compare the final rankings from the hierarchical Glicko-2 algorithms and the Glicko-2 algorithms to the world rankings produced by the ATP and WTA at the time of the final rating period, the 13th of January 2020. In general, the sum aggregation function produced a lower RMSE and MAE for both the ATP and WTA rankings than the mean aggregation function. This means that on average a male or females ranking from the hierarchical Glicko-2 algorithm and Glicko-2 algorithm will be closer to their ranking from the respective professional tennis bodies. However, the

number of players ranked in the top 500 by both algorithms without professional status is less for the mean aggregation function for both males and females. Having fewer veteran or junior players ranked ahead of professional players is important for the system since it would be rare that they would be better than a professional.

We can clearly see the need for grouping players when we compare the world rankings with the Glicko-2 algorithm's rankings. The issue that has emerged is that the algorithm cannot tell the difference between a top professional and a veteran or junior playing at a much lower level. The Glicko-2 algorithm only has access to set results and the players at the top of the rankings have similar win records, despite the level of opponent. We have opted not to show the rankings due to confidentiality. The hierarchical Glicko-2 algorithm produces better rankings than the Glicko-2 algorithm regardless of aggregation function, but there are still some improvements to be made to ensure that only the best players in the world are ranked so highly. We make some suggestions for improvements in the next chapter but for now we recommend using the hierarchical Glicko-2 algorithm with the mean aggregation function.

Table 4.2.: The root mean squared error (RMSE) and the mean absolute error (MAE) of the rankings from the hierarchical Glicko-2 algorithm and the Glicko-2 algorithm compared to the ATP and WTA rankings. ATP and WTA Top 500 is the number of players in the top 500 male and female rankings from the hierarchical Glicko-2 algorithm and the Glicko-2 algorithm without professional status.

Function	ATP			WTA		
	RMSE	MAE	Top 500	RMSE	MAE	Top 500
Sum (hierarchical Glicko-2)	500	291	77	342	220	104
Mean (hierarchical Glicko-2)	571	314	46	416	251	86
Sum (Glicko-2)	1868	1483	464	941	725	347
Mean (Glicko-2)	2010	1378	396	1058	729	288

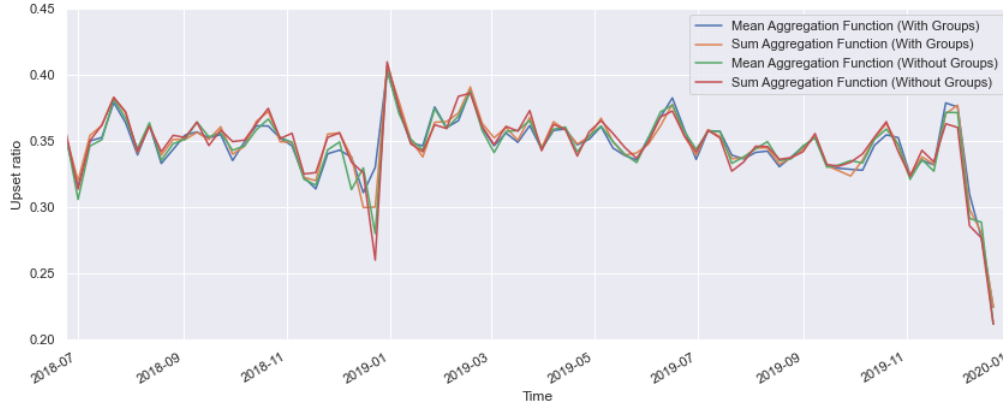


Figure 4.1.: Upset ratio of tennis set predictions based on ratings from the hierarchical Glicko-2 algorithm and the Glicko-2 algorithm using the mean aggregation function and the sum aggregation function.



Figure 4.2.: Log loss of tennis set predictions based on ratings from the hierarchical Glicko-2 algorithm and the Glicko-2 algorithm using the mean aggregation function and the sum aggregation function.

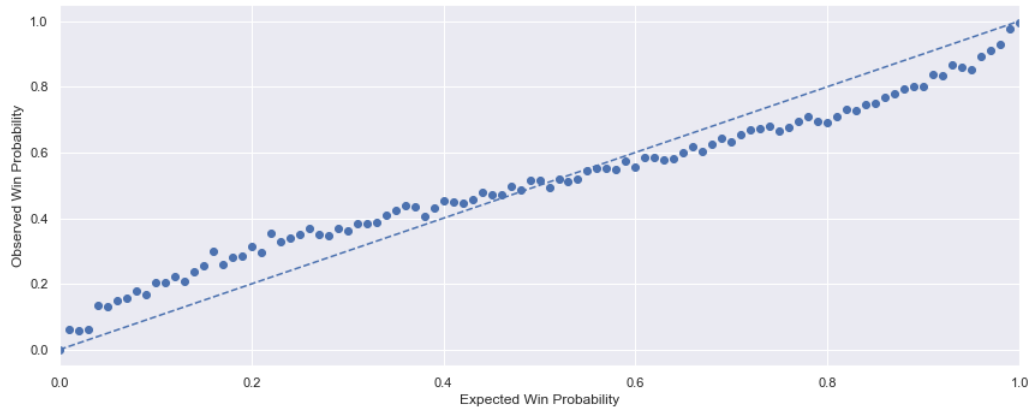


Figure 4.3.: The expected win probability versus the observed win probability based on the tennis set predictions from the hierarchical Glicko-2 algorithm using the sum aggregation function.

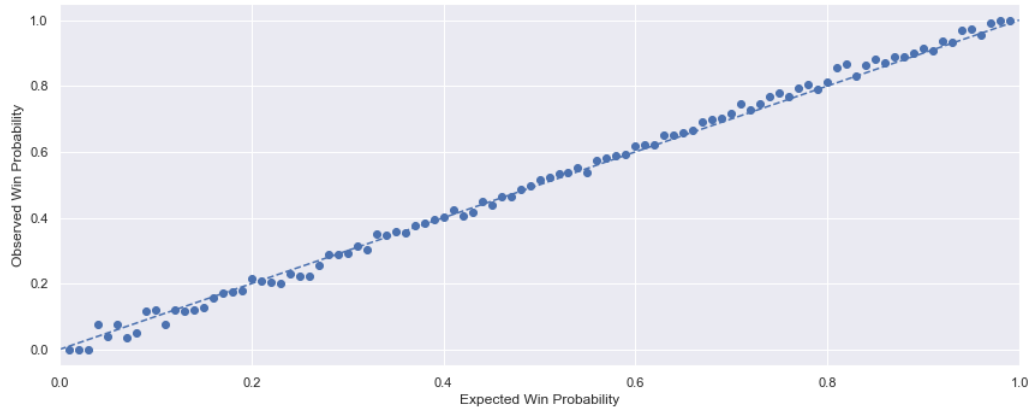


Figure 4.4.: The expected win probability versus the observed win probability based on the tennis set predictions from the hierarchical Glicko-2 algorithm using the mean aggregation function.

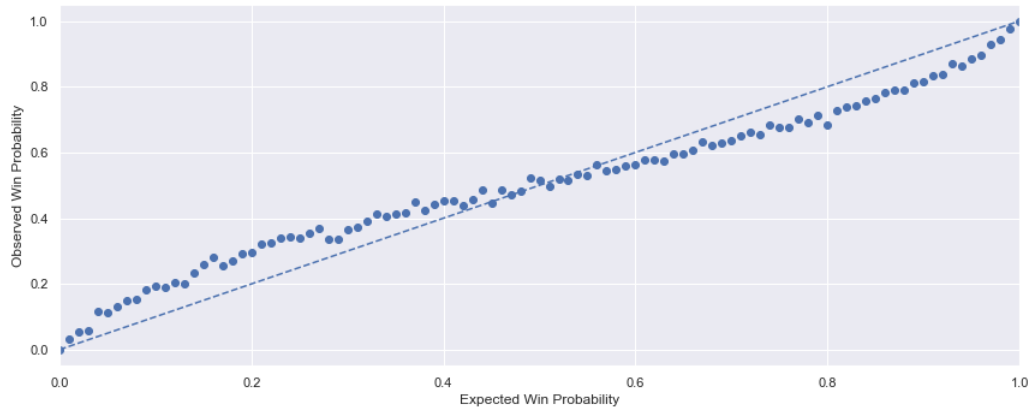


Figure 4.5.: The expected win probability versus the observed win probability based on the tennis set predictions from the Glicko-2 algorithm using the sum aggregation function.

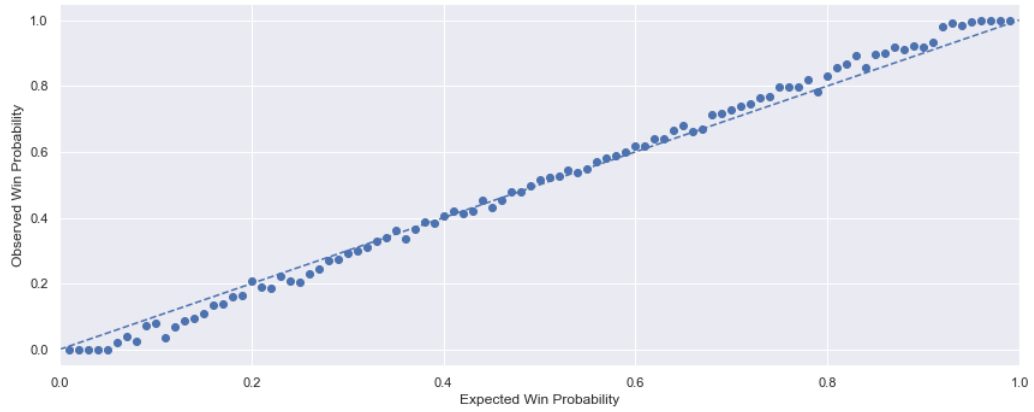


Figure 4.6.: The expected win probability versus the observed win probability based on the tennis set predictions from the Glicko-2 algorithm using the mean aggregation function.

5. Conclusion

In this chapter, we discuss the achievements made throughout the duration of this project and propose some future avenues of work for those working on the WTN initiative.

5.1. Achievements

When we started on this project with the ITF much research had been conducted on both singles tennis and doubles tennis. The work outlined in this report set about adapting and improving the hierarchical Glicko-2 algorithm for doubles tennis and building upon the concepts of a notional opponent that already existed.

Over 4 months we improved the concept of a notional opponent by removing the skill of a player from their opponent's skill. We decomposed the notional opponent into a player element and a group element, termed the notional player and notional group. For the mean and sum skill aggregation functions we computed the notional opponent of a player and of a group on a case-by-case basis. Lastly, we evaluated skill aggregation functions by means of simulation and real tennis data. The results of the comparison also showed the need for the hierarchical Glicko-2 algorithm over the Glicko-2 algorithm. Our conclusion is that the mean aggregation function is better suited for doubles tennis than the sum aggregation function and that the hierarchical Glicko-2 algorithm should be preferred to the Glicko-2 algorithm.

5.2. Future Work

5.2.1. Initialisation

The current version of the hierarchical Glicko-2 algorithm used for singles tennis makes use of an initialisation model for player ratings. This could be adapted for doubles tennis to accelerate the convergence of players relative skills to their true value and improve the current metrics we use to judge the algorithms performance. Factors such as age, gender and even professional status could be used as part of our prior beliefs for player skill.

5.2.2. Male and Female Group Alignment

Doubles tennis usually pits pairs of the same gender against each other while mixed doubles pairs males with females. Thus there is very little direct comparison between male and female players. Groups produced by the Leiden algorithm contain mostly male or female players. Aligning these groups correctly is of much importance for the WTN. This represents a unique challenge due to the lack of data, so some form of subjectivity may be required.

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A. Notional Opponent Skill Derivations

In this appendix we show how to derive results from chapter 3 regarding the skill of a notional opponent.

A.1. Mean

Glickman (1999) shows the log-likelihood of the Bayesian model that compares player i to player j can be approximated as

$$\ell(\gamma_i | \mathbf{s}) \approx \sum_{j=1}^M \sum_{k=1}^{K_j} \left(g(\phi_j^2) s_{jk} (\gamma_i - \mu_j) - \log(1 + \exp(g(\phi_j^2)(\gamma_i - \mu_j))) \right).$$

Using the mean skill aggregation function and letting j represent the notional opponent, then we can adapt this formula to get

$$\ell(\gamma_i | \mathbf{s}) \approx \sum_{j=1}^M \sum_{k=1}^{K_j} \left(g(\phi_j^2) s_{jk} \left(\frac{\gamma_i}{2} - \mu_j \right) - \log(1 + \exp(g(\phi_j^2) \left(\frac{\gamma_i}{2} - \mu_j \right))) \right).$$

Now the first and second derivatives of the log likelihood are,

$$\begin{aligned} \frac{\partial \ell}{\partial \gamma_i} &= \sum_{j=1}^M \sum_{k=1}^{K_j} \frac{g(\phi_j^2)}{2} \left(s_{jk} - \frac{1}{1 + \exp(-g(\phi_j^2) \left(\frac{\gamma_i}{2} - \mu_j \right))} \right), \\ \frac{\partial^2 \ell}{\partial \gamma_i^2} &= - \sum_{j=1}^M \sum_{k=1}^{K_j} \frac{g(\phi_j^2)^2}{2^2} \left(\frac{1}{1 + \exp(-g(\phi_j^2) \left(\frac{\gamma_i}{2} - \mu_j \right))} \right) \left(1 - \frac{1}{1 + \exp(-g(\phi_j^2) \left(\frac{\gamma_i}{2} - \mu_j \right))} \right). \end{aligned}$$

The posterior mean and posterior variance of γ_i can be expressed as

$$\begin{aligned} \phi_i^2 &= (\phi_i'^{-2} + \nu^{-2})^{-1}, \\ \mu_i &= \mu_i + \frac{\eta}{\phi_i'^{-2} + \nu^{-2}}, \end{aligned}$$

where $\nu^{-2} = -\frac{\partial^2 \ell}{\partial \gamma_i^2}$ and $\eta = \frac{\partial \ell}{\partial \gamma_i}$ both evaluated at $\gamma_i = \mu_i$. This follows the same process that Glickman used to derive the results for the original Glicko algorithm.

A.2. Maximum

Assume $\gamma_i \sim \mathcal{N}(\mu_i, \phi_i^2)$, $\gamma_j \sim \mathcal{N}(\mu_j, \phi_j^2)$ and $\gamma_i \perp\!\!\!\perp \gamma_j$.

$$\begin{aligned}\text{Cov}(\gamma_i, \max(\gamma_i, \gamma_j)) &= \mathbb{P}(\gamma_i > \gamma_j) \text{Cov}(\gamma_i, \gamma_i) + \mathbb{P}(\gamma_i \leq \gamma_j) \text{Cov}(\gamma_i, \gamma_j) \\ &= \phi_i^2 \Phi\left(\frac{\mu_j - \mu_i}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right)\end{aligned}$$

A.3. Minimum

Assume $\gamma_i \sim \mathcal{N}(\mu_i, \phi_i^2)$, $\gamma_j \sim \mathcal{N}(\mu_j, \phi_j^2)$ and $\gamma_i \perp\!\!\!\perp \gamma_j$.

$$\begin{aligned}\text{Cov}(\gamma_i, \min(\gamma_i, \gamma_j)) &= \mathbb{P}(\gamma_i \leq \gamma_j) \text{Cov}(\gamma_i, \gamma_i) + \mathbb{P}(\gamma_i > \gamma_j) \text{Cov}(\gamma_i, \gamma_j) \\ &= \phi_i^2 \Phi\left(\frac{\mu_i - \mu_j}{(\phi_i^2 + \phi_j^2)^{\frac{1}{2}}}\right)\end{aligned}$$